

API 610 Multistage Centrifugal Pumps - Technical Manual

Manufacturer Technical Guidance • Operational Boundaries & Maintenance Standards

1. Scope & Equipment Specification

This technical manual governs the operational limits, alignment tolerances, and preventive overhaul cycles for API 610 BB3 Heavy-Duty Multistage Centrifugal Pumps deployed in refinery fluid transport.

Design Pressure Rating: Class 600 RF. Casing Metallurgy: ASTM A216 WCB with 12% Chrome internals (API Material Class S-6).

Bearings: Hydrodynamic radial sleeve bearings with double-direction tilting pad thrust bearing.

2. Vibration Severity Standards (ISO 10816 / API 610)

Zone A (Newly commissioned machinery): < 2.3 mm/s RMS.

Zone B (Acceptable for unrestricted long-term operation): 2.3 to 4.5 mm/s RMS.

Zone C (Dissatisfactory - initiate maintenance plan): 4.5 to 7.1 mm/s RMS.

Zone D (Severe damage imminent - immediate trip): > 7.1 mm/s RMS.

Alarm threshold must be configured in DCS at 4.8 mm/s RMS; automatic trip threshold at 7.5 mm/s RMS.

3. Lube Oil System Operational Boundaries

Oil Reservoir Temperature: Maintain between 40°C and 55°C. Maximum allowable bearing drain temperature: 75°C.

Oil Supply Pressure to bearings: 1.5 to 2.2 bar(g). Cleanliness requirement: ISO 4406 code 17/15/12 maximum.

Lubricant grade: High-grade rust- and oxidation-inhibited (R&O) circulating oil ISO VG 46 or VG 68.

4. Overhaul Inspection Sequence

Check impeller ring diametral clearances against baseline: Replace wear rings if clearance exceeds 1.5x design.

Rotor dynamic balance: Balance to ISO 1940 Grade G1.0 or better.

Hydrostatic pressure test: Test casing at 1.5 times maximum allowable working pressure for minimum 30 minutes.